MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M26, ASSEMBLY

1. SCOPE

1.1 This specification covers the Thruster, Cartridge Actuated, M26, Assembly, loaded with Cartridge, Impulse, M259 for use in conjunction with other propellant actuated devices in aircrew emergency systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10551371	- Thruster, Cartridge Actuated, M26, Assembly

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

C9243197	- Container, Ammunition, Fiber for Thruster, Cartridge Actuated, M26
C9243198	- Box, Packing, Ammunition for Thruster, Cartridge Actuated, M26
IEL-10551371	- Index of Inspection Equipment List for Thruster, Cartridge Actuated, M26, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General - Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Torque - Torque requirements shall be as specified in Table I. No movement shall occur upon application of the breakaway torque.

Table I

Components	Seating or Bottoming (including A) (inch pounds)	Breakaway (inch pounds) (Minimum)
Cap Assy and Inside Tube Assy	300 ± 50	200
Sleeve and Outside Tube Assy	500 ± 50	400

3.3 Assembly

3.3.1 Initial Lock - There shall be no unlocking of the sleeve from the cap assy when a tensile load of 30 to 40 pounds is applied.

3.3.2 Residual magnetism - The thruster shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional

3.4.1 The thruster shall not fail to fire.

3.4.2 The thruster shall function without mechanical failure (see 6.2).

3.4.3 No load - The thruster, fired with no load on the outside tube, shall complete its stroke and the tubes shall not separate (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

3.4.4 Load - The thruster shall be mounted normal to a load having a mass moment of inertia of 7.2 slug ft^2 and shall rotate the load thru an arc of 90° when fired.

3.4.4.1 Velocity - The average velocity of the load during the first 5.75 inches of load rotation shall be 10.4 radians/sec. min.

3.4.4.2 Thrust - Maximum thrust shall not exceed 7000 pounds.

3.4.4.3 Ignition delay - Ignition delay shall not exceed 60 milliseconds (see 6.5).

3.4.5 Temperature - The thruster shall comply with all the ballistic requirements throughout the temperature range of minus 65° to plus 200°F.

3.4.6 Initiation - The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4 MS28741, @70° ± 5°F.

3.5 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects - All components shall be free from cracks, splits, perforations and any similar defect.

3.5.2 Burr - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter - No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions

4.2.1 Submission of product - Unless otherwise specified by the contracting officer, formation of lots, lot size and presentation of lots shall be in accordance with the "Submission of Products", provisions of MIL-STD-105.

4.2.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Minor		Major	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1,800	90	1	2	2	3
1801 - 10,000	200	2	3	3	4

4.2.2.1 Cartridges, Impulse, M259, Dwg. D10551387, covering a detail of Dwg. F10551371

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.2.2.2 Assembly, Drawing F10551371, less Shipping Cap

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Threads damaged	Visual
102. Foreign matter	Visual
103. Total length	Gage
MINOR	
None defined	

4.2.2.3 Assembly, Dwg. F10551371

CRITICAL	
1. Any part missing as detected by x-ray	X-ray
2. Improper assembly	X-ray
3. Metal defective	Visual
MAJOR	
101. Marking missing, incorrect or illegible	Visual
102. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Protective finish inadequate or defective	Visual
203. Shipping Cap missing	Visual
204. Location of labels incorrect	Gage

4.2.2.4 Fiber Container, Dwg. C9243197

CRITICAL	
None defined	
MAJOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.2.2.4 - Cont'd

Defect		Method of Inspection
MINOR		
201.	Gap between body and cover, max.	Manual
202.	Contents loose	Manual
203.	Metal ends loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min.	Visual
208.	Color or tape incorrect	Visual

4.2.2.5 Packing Box, Dwg. C9243198

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual & Visual
203.	Strapping missing, loose, broken or improperly engaged	Manual & Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing - The following tests shall be performed in accordance with 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.2.3.1 Breakaway torque - Each cap assembly and inside tube assembly and each sleeve and outside tube assembly shall be subjected to a breakaway torque test. Failure of any unit to comply with the requirements of 3.2 and Table I shall be cause for rejection.

4.2.3.2 Residual magnetism - Each thruster shall be subjected to the test for excessive residual magnetism. Failure of any thruster to comply with the requirement of 3.3.2 shall be cause for rejection of the thruster.

4.2.3.3 Initial Lock - Each thruster shall be subjected to the tensile load application. Failure of any thruster to comply with the requirement of 3.3.1 shall be cause for rejection of the unit.

4.2.3.4 Functional tests - The following tests shall be performed by the Government:

4.2.3.4.1 Ballistics - Thirty-three (33) units shall be tested to determine compliance with the requirements of 3.4.1 thru 3.4.4.3. The units shall be temperature conditioned as provided in Table III and 4.3.4.1.1. Failure of any unit to comply with the requirement or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

Table III

Sample Size	Method	Temperature
10	Load (See 3.4.4)	-65 to -70 F
10	Load (See 3.4.4)	+65 to +75 F
10	Load (See 3.4.4)	+200 to +205 F
1	No Load (See 3.4.3)	-65 to -70 F
1	No Load (See 3.4.3)	+65 to +75 F
1	No Load (See 3.4.3)	+200 to +205 F

4.2.2 Inspection equipment - The examination and tests shall be made using equipment in accordance with IEL-10551371.

4.3 Test methods and procedures

4.3.1 Breakaway torque - A torque wrench, capable of measuring the torque requirements as specified in Table I shall be used with an appropriate adapter to determine the adequacy of the joints.

4.3.2 Initial Lock - Upon completion of assembly, a tensile load of 30 to 40 pounds shall be applied between the sleeve and cap to determine that the initial lock shear pin is properly engaged. The load shall not be applied at a greater rate than .125 inches per minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.3.3 Residual magnetism - The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

4.3.4 Functional tests

4.3.4.1 Ballistics - The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting point. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperature. Test equipment and procedures shall be approved by the responsible technical agency prior to acceptance testing.

4.3.4.1.1 Temperature conditioning - A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A - Packing and marking shall be in accordance with Drawings C9243197 and C9243198.

6. NOTES

6.1 Ordering data - Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a designed function.

6.3 No load - Permanent yielding of the tubes shall not be cause for rejection when the thrusters are tested with no load on the outside tube.

6.4 Load - The expected performance of the thruster under conditions simulating those existing in aircraft is obtained when thrusters satisfactorily me conditions of the load firing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

6.5 "Ignition delay" - Is defined as the time lapse between the firing of the cartridge and the beginning of the continuous pressure rise in the body of the thruster.

Custodian:

Army - MU

Preparing activity:

Army - MU
Project No. 1377-A422

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.